

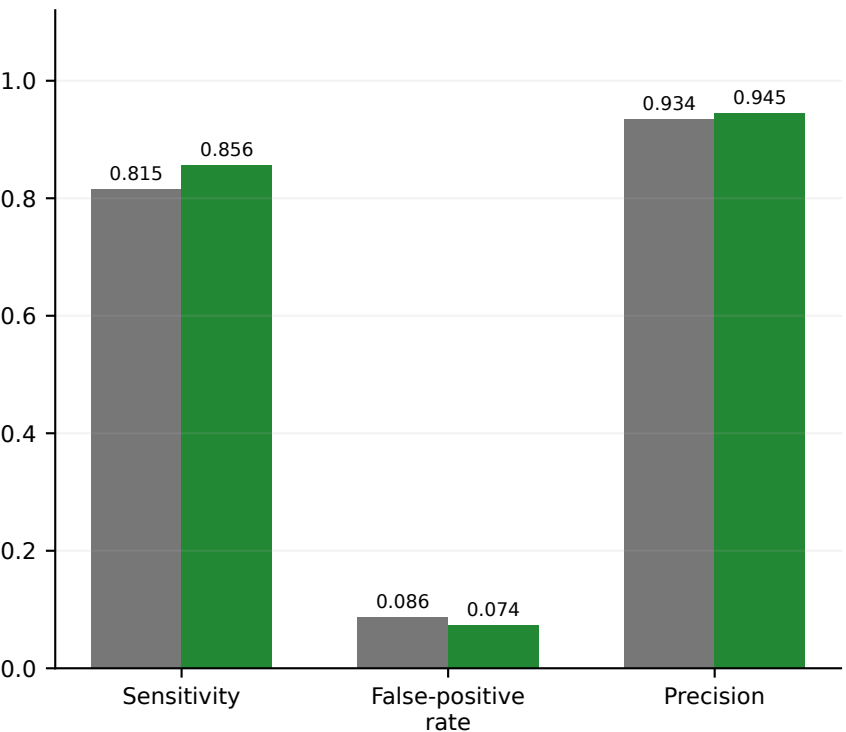
Frozen multi-k rule: fresh conditional assembly validation

A Predeclared validation design

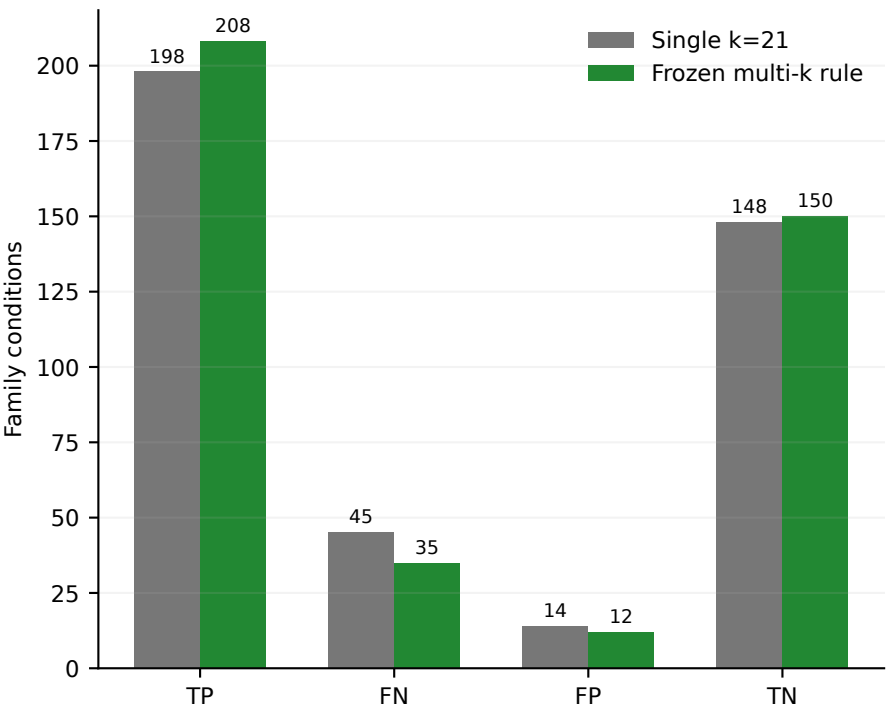
Fresh genomes	3	5201-5203
Coverage	3	1x, 5x, 20x
Read-error tiers	3	0, 0.1%, 1% substitutions
Assembly fractions	5	100%, 75%, 50%, 25%, 0%
Family conditions	405	243 positive + 162 control

Rule and hashes committed before execution; no held-out fitting

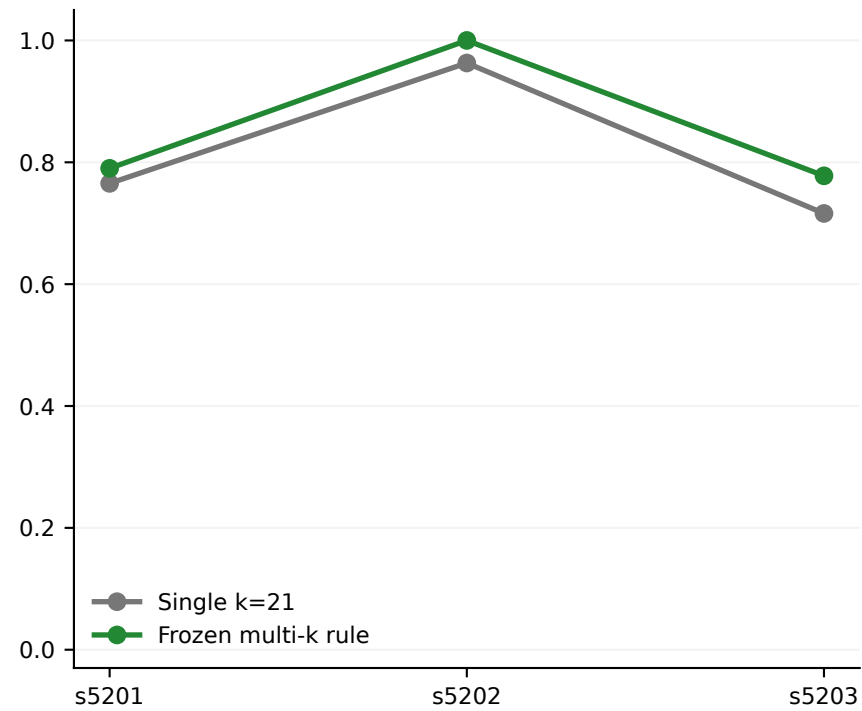
B Overall classification metrics



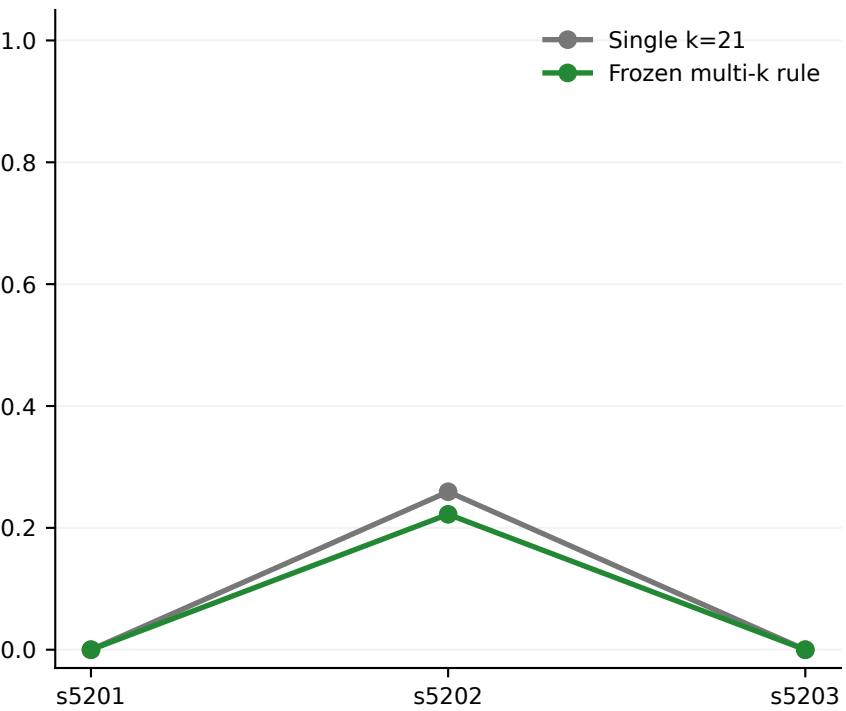
C Confusion counts



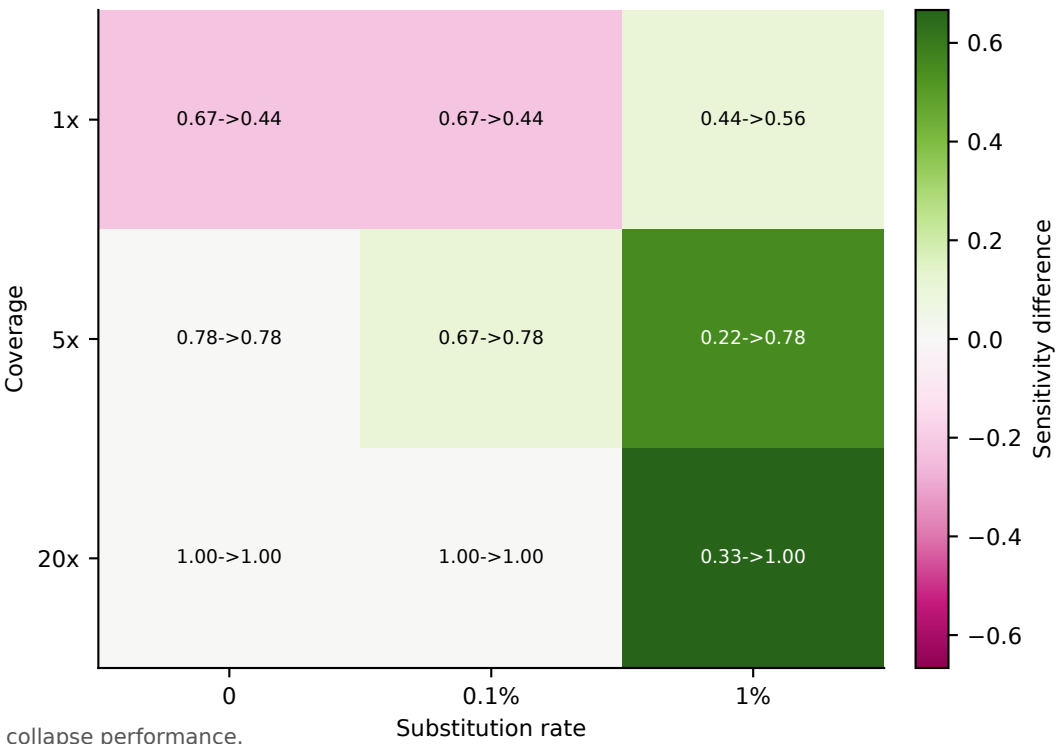
D Sensitivity by fresh genome



E False-positive rate by fresh genome



F Recall gain at 50% retention



Exact-copy known-catalogue simulations; improvement does not establish biological collapse performance.